

# Linear Regression

## HW 3

Read this before you start. Following these steps carefully will save you time at the end.

### 1. Environment

If you haven't already set up this course's environment:

```
git pull          # get the latest assignment files
uv sync           # install/update the course environment
```

Now you can open a notebook and select the course `.venv` as your kernel (VSCode: top-right kernel picker → Python Environments → the environment matching this repo's folder).

### 2. Answering questions

- **Written/math questions:** Write your answers in separate Markdown cells for each question and clearly denote which question number it is.
  - If you know  $\text{\LaTeX}$ , you can typeset math with `$$ ... $$` (e.g. `$$\hat{\beta} = (X^T X)^{-1} X^T y$$`).
  - If you don't know  $\text{\LaTeX}$ , that's completely fine, either:
    - \* write it in plain text (e.g. `beta_hat = (X^T X)^-1 X^T y`), or
    - \* solve it by hand on paper, take a clear, right-side-up, well-lit photo, and insert it directly into the Markdown cell:

```
![Q2 solution](q2_handwritten.jpg)
```

Keep the image file in the same folder as your notebook so it displays correctly.

- **Coding questions:** write your code in separate code cells for each question. Include any plots as cell output.
- Keep printed output reasonably short. Use `.head()` instead of printing full dataframes, and avoid excessively long console output between questions.

### 3. Before exporting

Re-run all code. For example, run **Kernel** → **Restart & Run All** (VS Code: the “Restart” button, then run all cells top to bottom) so that every output in your final PDF reflects your current code and not a stale run from earlier. Scroll through and confirm every plot actually rendered before moving on.

### 4. Export to PDF

Convert to pdf using nbconvert. If using VS Code: open the notebook → click the “...” menu in the notebook toolbar → **Export** → **PDF**.

(If that fails, see Troubleshooting below.)

## 5. If you did any part by hand on separate paper (not inserted as an image above)

- Scan or photograph those pages into a single PDF (any phone scanning app works, e.g. the Notes app on iPhone, or Google Drive's scan feature on Android).
- Export your notebook (coding + any typed answers) to its own PDF.
- Merge the two into **one final PDF**, in the same order as the assignment questions:
  - **Mac:** open one PDF in Preview, drag the other PDF's pages into the sidebar.
  - **Windows/other:** use a free tool like [PDFsam](#) or an online PDF merge tool.

## 6. Submit

Upload **one combined PDF** with all written answers, code, and plots, in question order, to Gradescope.

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## Troubleshooting

- **PDF export fails in VS Code:** run this from the terminal instead:

```
uv add --dev "nbconvert[webpdf]" playwright
uv run playwright install chromium
uv run jupyter nbconvert --to webpdf --allow-chromium-download solution.ipynb
```

- **Image doesn't show up in the exported PDF:** check that the image file is in the same folder as the notebook and that the filename in `![...]` matches exactly (case-sensitive).
- **ModuleNotFoundError when running code:** check the kernel selected in VS Code matches this course's environment, not a different Python install.

### Written Questions:

1. Set up the  $\mathbf{X}$  matrix and  $\beta$  vector for each of the following regression models (assume  $i = 1, \dots, 5$ ):
  - a.  $Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1}^2 + \varepsilon_i$
  - b.  $\sqrt{Y_i} = \beta_0 + \beta_1 X_{i1} + \beta_2 \log_{10} X_{i2} + \varepsilon_i$
2. A student stated: "Adding predictor variables to a regression model can never reduce  $R^2$ , so we should include all available predictor variables in the model." Comment.
3. Consider the multiple regression model:

$$Y_i = \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i \quad i = 1, \dots, n$$

where the  $\varepsilon_i$  are uncorrelated, with  $E\{\varepsilon_i\} = 0$  and  $\sigma^2\{\varepsilon_i\} = \sigma^2$ .

State the least squares criterion and derive the least squares estimators of  $\beta_1$  and  $\beta_2$ .

4. Consider the multiple linear regression model in the matrix form  $Y = X\beta + \epsilon$  with  $E(\epsilon) = 0$  and  $\text{Var}(\epsilon) = \sigma^2 I_n$ . Let  $\hat{\beta}$  be the least squares estimator of  $\beta$ ,  $\hat{Y} = X\hat{\beta}$  be the fitted values, and  $e = Y - \hat{Y}$  be the residuals. Express  $\text{Var}(\hat{\beta})$ ,  $\text{Var}(\hat{Y})$ , and  $\text{Var}(e)$  in terms of  $\sigma^2$  and  $X$ . Show your steps to obtain the expressions in detail.

### Coding questions:

1. In a small-scale regression study, the following data were obtained:

| $i$      | 1  | 2  | 3  | 4  | 5  | 6  |
|----------|----|----|----|----|----|----|
| $X_{i1}$ | 7  | 4  | 16 | 3  | 21 | 8  |
| $X_{i2}$ | 33 | 41 | 7  | 49 | 5  | 31 |
| $Y_i$    | 42 | 33 | 75 | 28 | 91 | 55 |

Assume that regression model with independent normal error terms is appropriate. Using matrix multiplication methods in Python, obtain (a)  $\hat{\beta}$ ; (b)  $\mathbf{e}$ ; (c)  $\mathbf{H}$ ; (d)  $SSR$ ; (e)  $SE(\hat{\beta})$ ; (f)  $\hat{Y}_h$  when  $X_{h1} = 10$ ,  $X_{h2} = 30$ ; and (g)  $SE(\hat{Y}_h)$  when  $X_{h1} = 10$ ,  $X_{h2} = 30$ .

**Do NOT** use Python packages to fit the model and read the output directly.

- The dataset “insurance.csv” contains information on the individuals’ medical costs billed by health insurance with the variables defined as below:

- age: age of primary beneficiary
- sex: insurance contractor gender, female, male
- bmi: Body mass index, providing an understanding of weights that are relatively high or low relative to height, objective index of body weight ( $kg/m^2$ ) using the ratio of height to weight
- children: Number of children covered by health insurance / Number of dependents
- smoker: Smoking
- region: the beneficiary’s residential area in the US, northeast, southeast, southwest, northwest.
- charges: Individual medical costs billed by health insurance

Regress “charges” against “age”, “bmi” and “children” (use the order:  $\text{charges} \sim \text{age} + \text{bmi} + \text{children}$ ) in Python.

- ONLY based on the model summary table (`model.summary()`), we have discussed several important things to look at when it comes to the model interpretation: Global F-test p-value,  $R^2$  and  $R_a^2$ , and t-test for individual coefficient. Comment on each values above from your model summary, what kind of conclusion you have about your model fit and significance of predictors so far?
  - Run `sm.stats.anova_lm(model, typ=1)` to obtain `sum_sq` of “bmi” in the ANOVA is a sequential sum of squares of “bmi” which is defined as  $SeqSS(bmi) = SSE_{reduced} - SSE_{full}$ . What is the reduced model and full model here? How do we interpret the p-value of “bmi” in this F-test?
  - Run `sm.stats.anova_lm(model, typ=2)` to obtain `sum_sq` of “bmi” in the anova is a partial sum of squares of “bmi” defined by  $PartialSS(bmi) = SSE_{reduced} - SSE_{full}$ . What is the reduced model and full model here? How do we interpret the p-value of “bmi” in this F-test?
- A commercial real estate company evaluates age ( $X_1$ ), operating expenses ( $X_2$ , in thousands of dollars), vacancy rate ( $X_3$ ), total square footage ( $X_4$ ) and rental rates ( $Y$ , in thousands of dollars) for commercial properties in a large metropolitan area in order to provide clients with quantitative information upon which to make rental decisions. The data are taken from 81 suburban commercial properties. (The data can be found in `property.txt`; The first column is  $Y$ , followed by  $X_1$ ,  $X_2$ ,  $X_3$ ,  $X_4$ .) Consider the multiple linear regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \epsilon_i, \quad i = 1, \dots, n, \quad \text{with } \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2).$$

- Fit the model in Python and produce the summary table.
- Compute the fitted values and residuals. Print the first 6 cases. Based on the residuals, give an estimate of  $\sigma^2$ .

- (c) Test whether  $\beta_2 = 0$  or not at 0.01 significance level. Interpret this coefficient in the real life terms of the problem.
- (d) Test  $H_0 : \beta_1 = \beta_2 = 0$ . vs.  $H_a : \text{either } \beta_1 \text{ or } \beta_2 \text{ not equal to } 0$ .
- (e) Consider a property with the following characteristics:  $X_1 = 4, X_2 = 10, X_3 = 0.1, X_4 = 80,000$ . Give a 90% prediction interval for the rental rate. The degrees of freedom for the t-distribution is  $n - p$ , and the standard error the prediction is given by

$$\text{SE}(\hat{Y}_h) = \sqrt{\hat{\sigma}^2 (1 + \mathbf{x}_h^T (X^T X)^{-1} \mathbf{x}_h)}$$